

FinTrust Digital Bank — Week 2 Data Analytics Project Documentation

AnalystLab Africa Experience Lab Internship Programme

Track: Data Analytics | Week 2: Analyse & Prepare

1. Week 1 Plan Applied

Week 1 planned data profiling, analytical questions, KPI definition and dashboard planning. Week 2 implements that plan through Excel data-quality assessment, SQL business analysis, Python EDA and Power BI preparation. The Week 1 join-integrity issue was carried forward as a gating item rather than ignored.

2. Approved Resources Used

The Week 2 brief identifies the approved customer CSV, transaction CSV, data dictionary, financial knowledge base, project brief and four-week roadmap. For this Data Analytics track, the customer/transaction exports and data dictionary were the relevant resources for the analytical outputs.

3. Data Quality Assessment

Check	Result	Status
Customers	33 records; 6 source fields in supplied export	Pass
Transactions	33 records; 6 fields	Pass
Missing values	0 in visible core fields	Pass
Duplicate IDs	0 customer IDs; 0 transaction IDs	Pass
Customer-ID join	1/33 matched; 32 unmatched	Investigate
Transaction_Status	Absent	Unavailable
Risk_Review_Flag	Absent	Unavailable
Potential IQR outliers	5 above NGN 53,278.94	Review

4. Key Analytical Results

Total transaction value is NGN 959,180.43 across 33 transactions. Mobile App is the most-used channel with 14 transactions (42.4%) and carries NGN 493,341.73. Everyday is the largest visible customer segment with 17 of 33 customers. Cash Withdrawal contributes the largest transaction value at NGN 395,660.05. The median transaction value is NGN 6,702.14, while the mean is NGN 29,066.07, indicating a right-skewed distribution.

5. Business Findings — Finding → Evidence → Business Meaning

Finding	Evidence	Business Meaning
Mobile App is the dominant transaction channel in this sample (42.4%) and NGN 493,341.73 in Digital-channel performance should be monitored using both transaction count and value	14 of 33 transactions (42.4%) and NGN 493,341.73	Digital-channel performance should be monitored using both transaction count and value
Channel volume and value tell different stories. USSD has 4 transactions but NGN 209,250.08, exceeding all other channels	USSD has 4 transactions but NGN 209,250.08, exceeding all other channels	Managers should be wary of relying on transaction count alone to assess channel importance
Cash Withdrawal drives the largest transaction value at NGN 395,660.05	Transaction totaling NGN 395,660.05	Cash movement is a major value driver in this sample and warrants close monitoring
Transaction values are strongly right-skewed. Mean NGN 29,066.07 vs median NGN 6,702.14; 5 IQR-based high-value observations	Mean NGN 29,066.07 vs median NGN 6,702.14; 5 IQR-based high-value observations	Mean and high-value observations are important so a few large transactions can skew the average
Customer-level transaction analysis is crucial. Only 1 of 33 transaction Customer_IDs matches the visible customer list	Only 1 of 33 transaction Customer_IDs matches the visible customer list	The customer relationship must be validated before customer segmentation

6. Testing / Evaluation Log

Work Completed	Test/Evaluation	Finding	Action/Improvement	Result
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Data profiling	Missing/duplicate/type checks	Core visible fields complete; IDs unique	Retain source rows	No row-level cleaning required
Join validation	Left join on Customer_ID	32/33 transactions unmatched	Block customer-level conclusions	Join issue explicitly documented
Outlier review	IQR rule	5 high-value observations above threshold	Flag for validation; do not delete automatically	Distribution remains intact
EDA	Cross-check summary tables vs results	Results consistent with source sample	Use count + value views	Ready for dashboard build
Dashboard KPI review	Field availability check	Status/risk KPIs unavailable	Exclude unsupported KPIs and show data points	Present only validated metrics

7. Important Decisions

Decision 1: Do not fabricate Transaction_Status or Risk_Review_Flag. Reason: the fields are not present in the supplied transaction export, and the Week 2 brief requires evidence-based analysis. Impact: success-rate and risk-review KPIs are marked unavailable. Decision 2: Do not force the customer/transaction relationship. Reason: only 1/33 transaction Customer_ID values match. Impact: transaction analysis remains transaction-level until source keys are validated. Decision 3: Do not remove high-value transactions solely because they are outliers. Reason: an outlier is not automatically an error. Impact: they are flagged for review while preserving the sample.

8. Week 2 Limitations and Risks

The visible transaction extract contains one date and a short time window. The customer export contains fewer fields than the data dictionary defines. Transaction status and risk-review fields are absent. The customer/transaction join is not reliable. These limitations constrain customer-level behaviour analysis and several requested KPIs.

9. Week 3 Focus

Validate the source relationship and obtain the missing transaction/customer fields; then build the Power BI dashboard using only supported measures, add validated customer-level analysis, and extend the visual narrative once the source data is complete.